

In the following exercises, all coordinates and components are given with respect to a right-oriented orthonormal frame $\mathcal{K}$.

5.1. In each of the following cases calculate the tensor product $\mathbf{v} \otimes \mathbf{w}$.

a) $\mathbf{v}(1, -1), \mathbf{w}(2, 0)$

b) $\mathbf{v}(1, 0, -1), \mathbf{w}(0, 1, 1)$

5.2. Consider the line $\ell : x + y - 1 = 0$. Determine the matrix forms of the parallel projection and of the parallel reflection in the line ℓ parallel to $\mathbf{v}(2, 1)$.

5.3. Determine the orthogonal projection of the point $A(2, 11, -5)$ on the plane $\pi : x + 4y - 3z + 7 = 0$ by determining the matrix form of the projection.

5.4. Determine the orthogonal reflection of the point $P(6, -5, 5)$ in the plane $\pi : 2x - 3y + z - 4 = 0$ by determining the matrix form of the reflection.

5.5. Project the point $P(4, 1, -2)$ on the z -axis parallel to the plane $\pi : x + y + z = 0$.

5.6. Write down the explicit 3×3 -matrix forms for parallel projections and reflections in $\mathbb{E}^3$.

5.7. Give the matrix form for the orthogonal reflections in the planes

$$\pi_1 : 3x - 4z = -1 \quad \text{and} \quad \pi_2 : 10x - 2y + 3z = 4 \quad \text{respectively.}$$

Determine the set of points which are fixed by the composition of these two reflections.

5.1. In each of the following cases calculate the tensor product $v \otimes w$.

a) $v(1, -1), w(2, 0)$

b) $v(1, 0, -1), w(0, 1, 1)$

a) $v \otimes w = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot [2 \ 0] = \begin{bmatrix} 2 & 0 \\ -2 & 0 \end{bmatrix}$ b) $v \otimes w = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \cdot [0 \ 1 \ 1] = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & -1 & -1 \end{bmatrix}$

5.2. Consider the line $\ell: x + y - 1 = 0$. Determine the matrix forms of the parallel projection and of the parallel reflection in the line ℓ parallel to $v(2, 1)$.

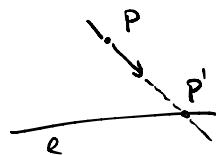
Method I

- take an arbitrary point $P(x_p, y_p)$
- construct line through P in direction of v

$$l_p: \begin{cases} x = x_p + 2t \\ y = y_p + t \end{cases}$$
- intersect ℓ with l_p to obtain the projection P' of P

$$l \cap l_p: \begin{cases} x = x_p + 2t \\ y = y_p + t \\ x + y - 1 = 0 \end{cases} \quad \begin{aligned} (x_p + 2t) + (y_p + t) - 1 &= 0 \\ \Rightarrow t &= \frac{1 - x_p - y_p}{3} \end{aligned}$$
- $\Rightarrow P' = l \cap l_p$ has coordinates $\left(x_p + 2 \frac{1 - x_p - y_p}{3}, y_p + \frac{1 - x_p - y_p}{3} \right)$
- separate the coordinates of the generic point to obtain

$$\begin{bmatrix} x_p \\ y_p \end{bmatrix} \mapsto \frac{1}{3} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_p \\ y_p \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$



Method II

apply the formula for the projection

$P_{r_{\ell, v}}(P) = \left(I_2 - \frac{v \cdot a^T}{v^T \cdot a} \right) P - \frac{a_3}{v^T \cdot a} v$ where $a = (1, 1)$
 $a_3 = -1$

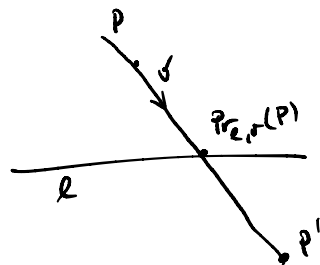
$\ell: x + y - 1 = 0$

$$= \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \right) P + \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_p \\ y_p \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

for the reflection:

Method I deduce it from the projection



$$\text{Ref}_{L,v}(P) = 2\text{Pr}_{L,v}(P) - P$$

$$\begin{aligned} &= 2 \left(\frac{1}{3} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_P \\ y_P \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right) - \begin{bmatrix} x_P \\ y_P \end{bmatrix} \\ &= \frac{1}{3} \left(\begin{bmatrix} 2 & -4 \\ -2 & 4 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \right) \begin{bmatrix} x_P \\ y_P \end{bmatrix} + \frac{2}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \frac{1}{3} \begin{bmatrix} -1 & -4 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x_P \\ y_P \end{bmatrix} + \frac{2}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \end{aligned}$$

Method II apply the formula for the reflection

$$\begin{aligned} \text{Ref}_{L,v}(P) &= \left(I_2 - 2 \frac{v \cdot a^T}{v^T \cdot a} \right) P - 2 \frac{a_3}{v^T \cdot a} v \\ &= \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{2}{3} \cdot \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \right) P + \frac{2}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \frac{1}{3} \begin{bmatrix} -1 & -4 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x_P \\ y_P \end{bmatrix} + \frac{2}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \end{aligned}$$

where as before
 $a = (1, 1)$
 $a_3 = -1$

5.3. Determine the orthogonal projection of the point $A(2, 11, -5)$ on the plane $\pi : x + 4y - 3z + 7 = 0$ by determining the matrix form of the projection.

$$\text{Pr}_{\pi}^{\perp}(P) = \left(I_3 - \frac{n n^T}{|n|^2} \right) P - \frac{a_4}{|n|^2} n \quad \text{where } n = \begin{bmatrix} 1 \\ 4 \\ -3 \end{bmatrix} \text{ and } a_4 = 7$$

$$n = \begin{bmatrix} 1 \\ 4 \\ -3 \end{bmatrix} \cdot [1 \ 4 \ -3] = \begin{bmatrix} 1 & 4 & -3 \\ 4 & 16 & -12 \\ -3 & -12 & 9 \end{bmatrix} \quad |n|^2 = 1 + 16 + 9 = 26$$

$$\text{Pr}_{\pi}^{\perp}(P) = \frac{1}{26} \left(\begin{bmatrix} 26 & 0 & 0 \\ 0 & 26 & 0 \\ 0 & 0 & 26 \end{bmatrix} - \begin{bmatrix} 1 & 4 & -3 \\ 4 & 16 & -12 \\ -3 & -12 & 9 \end{bmatrix} \right) P - \frac{7}{26} \begin{bmatrix} 1 \\ 4 \\ -3 \end{bmatrix} = \frac{1}{26} \begin{bmatrix} 25 & -4 & 3 \\ -4 & 10 & +12 \\ 3 & 12 & 17 \end{bmatrix} P - \frac{7}{26} \begin{bmatrix} 1 \\ 4 \\ -3 \end{bmatrix}$$

$$\text{Pr}_{\pi}^{\perp}(A) = \frac{1}{26} \begin{bmatrix} 25 & -4 & 3 \\ -4 & 10 & +12 \\ 3 & 12 & 17 \end{bmatrix} \begin{bmatrix} 2 \\ 11 \\ -5 \end{bmatrix} - \frac{7}{26} \begin{bmatrix} 1 \\ 4 \\ -3 \end{bmatrix} = \frac{1}{26} \begin{bmatrix} 50 & -44 & -15 & -7 \\ -8 & +110 & -60 & -28 \\ 6 & +132 & -85 & +21 \end{bmatrix} = \frac{1}{26} \begin{bmatrix} -16 \\ 14 \\ 74 \end{bmatrix} = \frac{1}{13} \begin{bmatrix} -8 \\ 7 \\ 37 \end{bmatrix}$$

you may also deduce the matrix form from first principles:

A N.V for π is $n(1, 4, -3)$

So the orthogonal projection is a projection along lines parallel to n

Let $P(x_0, y_0, z_0) \in \mathbb{E}^3$ and consider the line $\ell: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 4 \\ -3 \end{bmatrix} = \begin{bmatrix} x_0 + t \\ y_0 + 4t \\ z_0 - 3t \end{bmatrix}$

The orthogonal projection of P on π is $P' = \pi \cap \ell$

$$\pi \cap \ell: (x_0 + t) + 4(y_0 + 4t) - 3(z_0 - 3t) + 7 = 0$$

$$x_0 + t + 4y_0 + 16t - 3z_0 + 9t + 7 = 0$$

$$\Rightarrow t = \frac{1}{26} (-x_0 - 4y_0 + 3z_0 - 7)$$

one can also apply
the formula
to get this matrix

$$\Rightarrow P' = \begin{bmatrix} x_0 + \frac{-x_0 - 4y_0 + 3z_0 - 7}{26} \\ y_0 + 4 \frac{-x_0 - 4y_0 + 3z_0 - 7}{26} \\ z_0 - 3 \frac{-x_0 - 4y_0 + 3z_0 - 7}{26} \end{bmatrix} = \frac{1}{26} \begin{bmatrix} 25 & -4 & 3 \\ -4 & 10 & 12 \\ 3 & 12 & 17 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} - \frac{7}{26} \begin{bmatrix} 1 \\ 4 \\ -3 \end{bmatrix}$$

5.4. Determine the orthogonal reflection of the point $P(6, -5, 5)$ in the plane $\pi: 2x - 3y + z - 4 = 0$ by determining the matrix form of the reflection.

Let π be the plane and let $Q(x_0, y_0, z_0) \in \mathbb{E}^3$

A normal vector for π is $n(2, -3, 1)$

The orthogonal reflection in π is the reflection in π along lines parallel to n

$$\text{Ref}_{\pi}^{\perp}(Q) = \text{Ref}_{\pi, n}(Q) = \left(\text{Id}_3 - 2 \frac{n \cdot a^t}{n^t \cdot a} \right) Q - 2 \frac{-4}{n^t \cdot a} n$$

where $a = n$, so

$$n \cdot a^t = n \cdot n^t = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -6 & 2 \\ -6 & 9 & -3 \\ 2 & -3 & 1 \end{bmatrix}$$

$$n^t \cdot a = n^t \cdot n = \begin{bmatrix} 2 & -3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} = 4 + 9 + 1 = 14$$

$$\text{So } \left[\text{Ref}_{\pi}^{\perp}(Q) \right] = \frac{1}{14} \begin{bmatrix} 6 & 12 & -4 \\ 12 & -4 & 6 \\ -4 & 6 & 12 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + \frac{4}{7} \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$$

$$\text{In particular } \text{Ref}_{\pi}^{\perp}(P) = \begin{bmatrix} -2 \\ 7 \\ 1 \end{bmatrix}$$

5.5. Project the point $P(4, 1, -2)$ on the z -axis parallel to the plane $\pi: x + y + z = 0$.

Let P' be the projected point. Then $P' = Oz \cap \pi'$ where

π' is a plane passing through P and parallel to π

$$\Rightarrow \pi': (x-4) + (y-1) + (z+2) = 0 \Leftrightarrow \pi': x + y + z - 3 = 0$$

$$\text{So } P': \begin{cases} x + y + z - 3 = 0 \\ x = y = 0 \text{ (z-axis)} \end{cases} \Rightarrow z = 3 \Rightarrow P'(0, 0, 3)$$

5.6. Write down the explicit 3×3 -matrix forms for parallel projections and reflections in $\mathbb{E}^3$.

• We show this for projections in hyperplanes (the other case is similar)
 $\downarrow$
 $H: ax + by + cz + d = 0$ (a plane)
 $\downarrow$
 along $v = v(v_x, v_y, v_z)$

• the matrix form

$$P_{H,v}(P) = \left(I_n - \frac{v \cdot a^T}{v^T \cdot a} \right) P - \frac{a_{n+1}}{v^T \cdot a} v \quad \text{which in our case becomes}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto \left[\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \frac{1}{av_x + bv_y + cv_z} \begin{pmatrix} v_x a & v_x b & v_x c \\ v_y a & v_y b & v_y c \\ v_z a & v_z b & v_z c \end{pmatrix} \right] \begin{bmatrix} x \\ y \\ z \end{bmatrix} - \frac{d}{av_x + bv_y + cv_z} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$$

$$= \frac{1}{av_x + bv_y + cv_z} \begin{bmatrix} bv_y + cv_z & v_x b & v_x c \\ v_y a & av_x + cv_z & v_y c \\ v_z a & v_z b & av_x + bv_y \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} - \frac{d}{av_x + bv_y + cv_z} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$$

5.7. Give the matrix form for the orthogonal reflections in the planes

$$\pi_1: 3x - 4z = -1 \quad \text{and} \quad \pi_2: 10x - 2y + 3z = 4 \quad \text{respectively.}$$

Determine the set of points which are fixed by the composition of these two reflections.

$$\bullet \text{Ref}_H^\perp(P) = \left(I_3 - 2 \frac{a a^T}{|a|^2} \right) P - 2 \frac{a_4}{|a|^2} a$$

and we calculate this expression in both cases

• The set of fixed points is the intersection of the two planes.

$$R_2(R_1(P)) = P \Leftrightarrow R_1(P) = R_2(P) =: Y$$

if $Y \neq P$ then $\overrightarrow{PY}$ is a normal vector to both π_1 and π_2 which are not parallel $\Rightarrow R_1(P) = R_2(P) = P$